

## Problem Set 3

### ECON 2010 – Macroeconomics

### Spring 2019

Due at the start of class on March 6. Please complete all problems in this packet.

Note that this problem set will work the same as the first problem set and all problem sets. Work on a separate sheet of paper on your own, and we will correct as much of the problem set together in class as we can.

Name:

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#### Problem set instructions.

1. **Do not write on this problem set sheet** until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
  2. Write your final answers to each question on a separate sheet of paper.
  3. Write your final answers in pen, if possible.
  4. Ensure that your name appears on this problem set and on the separate sheet(s) of paper with your answers, and staple your answers to the back of this problem set.
  5. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
  6. Note that not all questions have *one* right answer. Some questions might have many correct answers. I try to indicate this.
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#### Problem 1 – Gross Domestic Product in Brazil (26 points)

The following table presents information on a simplified economy like Brazil's. Brazil is a diverse economy that produces many many goods and services. Some of the most important are agricultural — like beef, sugar, soybeans, and orange juice. Others are manufactured goods like airplanes, shoes, and cement. Yes, airplanes – the state-owned company Embraer produces small aircraft used as private planes and commuter planes for regional flights. Major airlines like Delta, United, etc. have Embraers in their fleets. And, of course, then there are also a multitude of services that the economy produces as well.

We'll conduct our calculations on an extremely simplified set of Brazil's goods described in the table below. While Brazil produces millions of tons of soybeans and oranges, we'll work with smaller numbers to make our calculations more manageable. Similarly, we'll remove some zeros from the prices of the planes. They're millions of dollars, but we don't need that complication.

Note 1: For each of the question, use the principle of *all else equal* (a.k.a. ceteris parabis, holding other factors fixed) to conduct your analysis and answer the question.

Note 2: **Whenever relevant, use 2015 as the base year for prices.**

Remember, you can use a calculator, but show how you set up the problem to get full points!

Table 1: Production and prices in an economy similar to Brazil

| Year | Quantity Produced       |                      |                             | Price in USD          |                    |                           |
|------|-------------------------|----------------------|-----------------------------|-----------------------|--------------------|---------------------------|
|      | Soybeans<br>(mil. tons) | Sugar<br>(mil. tons) | Orange Juice<br>(mil. tons) | Soybeans<br>(per ton) | Sugar<br>(per ton) | Orange Juice<br>(per ton) |
| 2015 | 115.0                   | 33.0                 | 1.1                         | \$390                 | \$60               | \$120                     |
| 2016 | 115.0                   | 32.0                 | 1.0                         | \$415                 | \$60               | \$135                     |
| 2017 | 119.0                   | 34.0                 | 1.2                         | \$430                 | \$65               | \$140                     |

(a) What is nominal GDP in 2015? **(2 points)**

(b) What is nominal GDP in 2016? **(2 points)**

(c) What is nominal GDP in 2017? **(2 points)**

(d) What is real GDP in 2015? **(2 points)**

(e) What is real GDP in 2016? **(2 points)**

(f) What is real GDP in 2017? **(2 points)**

(g) Assume that the population of Brazil is 100 in 2015. (It definitely isn't, but we're dealing with a simplified version of the economy to make our calculations reasonable.) What is the *nominal* per capita GDP in 2015? **(2 points)**

(h) Use 2015 population as given above. What is the *real* per capita GDP in 2015? **(2 points)**

(i) Assume that the population of Brazil is 120 in 2016. (It definitely isn't, but we're dealing with a simplified version of the economy to make our calculations reasonable.) What is the *nominal* per capita GDP in 2016? **(2 points)**

(j) Use 2016 population as given above. What is the *real* per capita GDP in 2016? **(2 points)**

- (k) Assume that the population of Brazil is 130 in 2017. (Again, it definitely isn't, but we're dealing with a simplified version of the economy to make our calculations reasonable.) What is the *nominal* per capita GDP in 2017? **(2 points)**
- (l) Use 2017 population as given above. What is the *real* per capita GDP in 2017? **(2 points)**
- (m) Imagine your friend looked at the numbers in the table and said: "Wow, Brazil's economy is bigger in 2017 as compared to 2015." Briefly describe the way(s) in which your friend is right, and the way(s) in which your friend is wrong. **(2 points)**

## Problem 2 (28 points)

In this problem, consider the town of Springfield, USA. There are ten people who live in Springfield (and one dog, but let's not consider Santa's Little Helper for this problem). The table below was compiled by the US government and gives you the age and employment status of the ten residents.

The US government obtained information for column 3 ("Has a job?") by calling each of the residents of Springfield and asking them if they had a job. The information for column 4 ("Is looking for a job?") was obtained by asking each individual whether they were actively looking for a job. Slightly different from usual, the interviewer asked everyone whether they were looking for a job and whether they had a job or not.

- (a) Using the information in the table and the definition of the unemployment rate, calculate the unemployment rate for Springfield. **(4 points)**

Table 2: Age and employment status of the residents of Springfield, USA

| Name             | Age | Has a job? | Is looking for a job? |
|------------------|-----|------------|-----------------------|
| Maggie           | 1   | No         | No                    |
| Lisa             | 7   | No         | No                    |
| Bart             | 9   | No         | No                    |
| Homer            | 45  | Yes        | No                    |
| Marge            | 44  | No         | No                    |
| Mo               | 56  | Yes        | No                    |
| Mr. Burns        | 102 | Yes        | No                    |
| Barney           | 42  | No         | Yes                   |
| Grampa           | 79  | No         | No                    |
| Krusty the Clown | 60  | Yes        | Yes                   |

(b) For each of the following changes, indicate whether the unemployment rate would *increase*, *decrease*, or *remain the same*. You do not need to calculate any change in the unemployment rate, but you may find it helpful to do so. Do not consider the changes cumulatively — that is, for each change, consider *only* that change relative to the information in the table. Do not consider a given change in addition to the previous changes. For each of the statements, use the principle of *all else equal* (a.k.a. *ceteris parabis*, holding other factors fixed) to conduct your analysis. **(6 points total)**

(1) Lisa decides to start looking for work as a professor of astrophysics. (Yeah, she's that smart.) But she hasn't yet found a job. **(1 point)**

(2) Homer loses his job at the nuclear power plant after spilling jelly from a donut in his control panel and nearly causing a meltdown. He's perfectly happy taking salsa lessons and drinking at a local bar, so he doesn't look for additional work. **(1 point)**

(3) Krusty the Clown has been looking for a new job in the movie industry (he only does TV at the moment), but gives up on his search and stops looking for new work. **(1 point)**

(4) Marge decides to hire someone to clean the house and teaches her husband Homer how to cook dinner when he gets off work, and sets out to find a job as a hairdresser for blue haired women. She immediately finds and starts a hairdressing job. **(1 point)**

(5) Mr. Burns is officially declared dead when the US government determines that too many of his organs are plastic or animal parts. The government no longer considers him part

of Springfield's population. **(1 point)**

(6) Both Grampa and Barney get part time jobs as checkers at the Kwik-E-Mart convenience store, but Barney keeps looking for full time work. **(1 point)**

(c) Using the information in the table and the definition of the labor market participation rate, calculate the labor market participation rate for Springfield. **(4 points)**

(d) For each of the following changes, indicate whether the labor force participation rate would *increase*, *decrease*, or *remain the same*. You do not need to calculate any change in the unemployment rate, but you may find it helpful to do so. Do not consider the changes cumulatively — that is, for each change, consider *only* that change relative to the information in the table. Do not consider a given change in addition to the previous changes. For each of the statements, use the principle of *all else equal* (a.k.a. *ceteris parabis*, holding other factors fixed) to conduct your analysis. **(6 points total)**

(1) Lisa decides to start looking for work as a professor of astrophysics. (Yeah, she's that

smart.) But she hasn't yet found a job. **(1 point)**

(2) Homer loses his job at the nuclear power plant after spilling jelly from a donut in his control panel and nearly causing a meltdown. He's perfectly happy taking salsa lessons and drinking at a local bar, so he doesn't look for additional work. **(1 point)**

(3) Krusty the Clown has been looking for a new job in the movie industry (he only does TV at the moment), but gives up on his search and stops looking for new work. **(1 point)**

(4) Marge decides to hire someone to clean the house and teaches her husband Homer how to cook dinner when he gets off work, and sets out to find a job as a hairdresser for blue haired women. She immediately finds and starts a hairdressing job. **(1 point)**

(5) Mr. Burns is officially declared dead when the US government determines that too many of his organs are plastic or animal parts. The government no longer considers him part of Springfield's population. **(1 point)**

(6) Both Grampa and Barney get part time jobs as checkers at the Kwik-E-Mart convenience store, but Barney keeps looking for full time work. **(1 point)**

(e) An old issue of the Springfield Inquirer newspaper from last year notes that the employment

rate between two months dropped from 8% to 6%. The columnist claims that this is unambiguously good news for the Springfield economy. As a well-trained student of economics, you want to think about that claim. Give **two** different reasons why the unemployment rate might have fallen, one of which is most people would consider good news, and one of which people would probably consider bad news. Note that your reasons don't need to involve specific people or situations, though they can if that is helpful. **(6 points total, 3 points each reason)**

- (f) Write your own normative statement about the inclusion of those that are only employed part time in the calculation of the unemployment rate. Limit yourself to *one* sentence — *a* statement. You will not be graded on the normative content of your statement, but instead whether you come up with a statement that is normative as opposed to positive. **(2 points)**

### **Problem 3 — Practice with Shifting Market Supply and Market Demand Curves (38 points)**

Consider the following labor market: plumbers. For simplicity, we'll consider only the labor market in one city, let's say Jackson, TN. And, as we always do, we'll assume that all plumbers in this market have the exact same skills and are, in effect, interchangeable. In this market, further assume that plumbers are all individuals who work for themselves (are self-employed), and those who purchase the labor of plumbers are homeowners who need plumbing work done on their houses.

In this problem you'll consider generic market supply and market demand curves. By generic, I

simply mean that I won't give you numbers to plot the specific points on the curves. Even without numbers to plot them, you know something about the shape of each of these curves (the first two questions below). And even without specific points, you can answer questions about how a generic curve would shift (*if at all*) given a particular change.

Apply the principle of *ceteris parabis* and hold all else equal when considering the changes in the following questions.

- (a) In the market for plumbers, which curve is upward sloping: market supply or market demand? In your own words, explain briefly why it is upward sloping. **(4 points)**
  
- (b) In the market for plumbers, which curve is downward sloping: market supply or market demand? In your own words, explain briefly why it is downward sloping. **(4 points)**
  
- (c) Consider only the following. The wage in the market for plumbers increases from \$18 to \$20 per hour. Given this change, does either the market supply or market demand curve shift? You can answer: *market supply*, *market demand*, *both* or *neither*. **(4 points)**
  
- (d) Consider the same change as in (c) above. In which direction does/do the curves shift, if at all. For each curve you can answer either: *right* or *left*. Or you can say *neither curve shifts*. **(2 points)**
  
- (e) Consider your answer in part (d) above. In your own words, explain briefly *why* the curves have or have not shifted as you indicated they would. Be careful with your explanation. Remember that an explanation is not simply a restatement of what happened. Think about

what the curves represent as this should help you to offer reasons *why*. **(4 points)**

- (f) Consider only the following. The state imposes a plumber certification that requires plumbers to take a written and practical exam in order to work as plumbers. The certification costs \$60 for plumbers to obtain, along with a few hours of time. Given this change, does either the market supply or market demand curve shift? You can answer: *market supply, market demand, both* or *neither*. **(4 points)**
- (g) Consider the same change as in (f) above. In which direction does/do the curves shift, if at all? For each curve you can answer either: *right* or *left*. Or you can say *neither curve shifts*. **(2 points)**
- (h) Consider your answer in part (h) above. In your own words, explain briefly *why* the curves have or have not shifted as you indicated they would. Be careful with your explanation. Remember that an explanation is not simply a restatement of what happened. Think about what the curves represent as this should help you to offer reasons *why*. **(4 points)**
- (i) Consider only the following. Software developers design a new app for your phone that walks you through plumbing repair yourself. It is less expensive to buy the app and spend your time doing many plumbing repairs yourself than hiring a plumber. Given this change, does either the market supply or market demand curve shift? You can answer: *market supply, market demand, both* or *neither*. **(4 points)**

(j) Consider the same change as in (i) above. In which direction does/do the curves shift, if at all? For each curve you can answer either: *right* or *left*. Or you can say *neither curve shifts*. **(2 points)**

(k) Consider your answer in part (j) above. In your own words, explain briefly *why* the curves have or have not shifted as you indicated they would. Be careful with your explanation. Remember that an explanation is not simply a restatement of what happened. Think about what the curves represent as this should help you to offer reasons *why*. **(4 points)**

### Correcting Your Problem Set (8 points)

You'll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others.